MATH 20250 | **HW5** | April 21, 2023
Prof. Lanier | | due: Monday, May 1, 2023

HW5

Submit 8 problems total from the union of *LADW* **Ch 4.1-2** (copied below) and the supplementary problems.

Please label all *LADW* problems by "Chapter.Section.Exercise," e.g. 1.3.4. Label the supplementary problems as they're labeled on the handout. Please make a note about any collaboration or use of outside resources.

Please put a "G" in front of the four problems you would like graded for correctness. Additionally, put an "F" in front of any of the problems graded for completion on which you would like written feedback from the grader.

Suggested *LADW* problems: 4.1.1, 4.1.6, 4.1.10, 4.1.11, 4.2.3, 4.2.8, 4.2.9, 4.2.12

Do not submit any of the problems marked with an X.

Section 4.1

Exercises.

1.1. True or false:

a) Every linear operator in an n-dimensional vector space has n distinct eigenvalues;
b) If a matrix has one eigenvector, it has infinitely many eigenvectors;
c) There exists a square real matrix with no real eigenvalues;
d) There exists a square matrix with no (complex) eigenvectors;
e) Similar matrices always have the same eigenvalues;
f) Similar matrices always have the same eigenvectors;
g) A non-zero sum of two eigenvectors of a matrix A is always an eigenvector;
h) A non-zero sum of two eigenvectors of a matrix A corresponding to the same eigenvalue λ is always an eigenvector.

$\boxed{\text{X}}$ **1.2.** Find characteristic polynomials, eigenvalues and eigenvectors of the following matrices:

$$\begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix}, \quad \begin{pmatrix} 2 & 1 \\ -1 & 4 \end{pmatrix}, \quad \begin{pmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{pmatrix}.$$

1.3. Compute eigenvalues and eigenvectors of the rotation matrix

$$\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}.$$

Note, that the eigenvalues (and eigenvectors) do not need to be real.

$\boxed{\text{X}}$ **1.4.** Compute characteristic polynomials and eigenvalues of the following matrices:

$$\begin{pmatrix} 1 & 2 & 5 & 67 \\ 0 & 2 & 3 & 6 \\ 0 & 0 & -2 & 5 \\ 0 & 0 & 0 & 3 \end{pmatrix}, \quad \begin{pmatrix} 2 & 1 & 0 & 2 \\ 0 & \pi & 43 & 2 \\ 0 & 0 & 16 & 1 \\ 0 & 0 & 0 & 54 \end{pmatrix}, \quad \begin{pmatrix} 4 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 2 & 4 & e & 0 \\ 3 & 3 & 1 & 1 \end{pmatrix},$$
$$\begin{pmatrix} 4 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 \\ 3 & 3 & 1 & 1 \end{pmatrix}.$$

Do not expand the characteristic polynomials, leave them as products.

1.5. Prove that eigenvalues (counting multiplicities) of a triangular matrix coincide with its diagonal entries

1.6. An operator A is called *nilpotent* if $A^k = \mathbf{0}$ for some k. Prove that if A is nilpotent, then $\sigma(A) = \{0\}$ (i.e. that 0 is the only eigenvalue of A).

1.7. Show that characteristic polynomial of a block triangular matrix

$$\begin{pmatrix} A & * \\ \mathbf{0} & B \end{pmatrix},$$

where A and B are square matrices, coincides with $\det(A - \lambda I)\det(B - \lambda I)$. (Use Exercise 3.11 from Chapter 3).

1.8. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be a basis in a vector space V. Assume also that the first k vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ of the basis are eigenvectors of an operator A, corresponding to an eigenvalue λ (i.e. that $A\mathbf{v}_j = \lambda\mathbf{v}_j$, $j = 1, 2, \dots, k$). Show that in this basis the matrix of the operator A has block triangular form

$$\begin{pmatrix} \lambda I_k & * \\ \mathbf{0} & B \end{pmatrix},$$

where I_k is $k \times k$ identity matrix and B is some $(n-k) \times (n-k)$ matrix.

1.9. Use the two previous exercises to prove that geometric multiplicity of an eigenvalue cannot exceed its algebraic multiplicity.

1.10. Prove that determinant of a matrix A is the product of its eigenvalues (counting multiplicities).

Hint: first show that $\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\dots(\lambda_n - \lambda)$, where $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigenvalues (counting multiplicities). Then compare the free terms (terms without λ) or plug in $\lambda = 0$ to get the conclusion.

1.11. Prove that the trace of a matrix equals the sum of eigenvalues in three steps. First, compute the coefficient of λ^{n-1} in the right side of the equality

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\dots(\lambda_n - \lambda).$$

Then show that $\det(A - \lambda I)$ can be represented as

$$\det(A - \lambda I) = (a_{1,1} - \lambda)(a_{2,2} - \lambda)\dots(a_{n,n} - \lambda) + q(\lambda)$$

where $q(\lambda)$ is polynomial of degree at most $n - 2$. And finally, comparing the coefficients of λ^{n-1} get the conclusion.

Section 4.2

Exercises.

2.1. Let A be $n \times n$ matrix. True or false:

a) A^T has the same eigenvalues as A.
b) A^T has the same eigenvectors as A.
c) If A is is diagonalizable, then so is A^T.

Justify your conclusions.

2.2. Let A be a square matrix with real entries, and let λ be its complex eigenvalue. Suppose $\mathbf{v} = (v_1, v_2, \dots, v_n)^T$ is a corresponding eigenvector, $A\mathbf{v} = \lambda\mathbf{v}$. Prove that the $\overline{\lambda}$ is an eigenvalue of A and $A\overline{\mathbf{v}} = \overline{\lambda}\overline{\mathbf{v}}$. Here $\overline{\mathbf{v}}$ is the complex conjugate of the vector $\mathbf{v}$, $\overline{\mathbf{v}} := (\overline{v}_1, \overline{v}_2, \dots, \overline{v}_n)^T$.

2.3. Let

$$A = \begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix}.$$

Find A^{2004} by diagonalizing A.

2.4. Construct a matrix A with eigenvalues 1 and 3 and corresponding eigenvectors $(1, 2)^T$ and $(1, 1)^T$. Is such a matrix unique?

$\boxed{\text{X}}$ **2.5.** Diagonalize the following matrices, if possible:

a) $\begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$.
b) $\begin{pmatrix} -1 & -1 \\ 6 & 4 \end{pmatrix}$.
c) $\begin{pmatrix} -2 & 2 & 6 \\ 5 & 1 & -6 \\ -5 & 2 & 9 \end{pmatrix}$ ($\lambda = 2$ is one of the eigenvalues)

2.6. Consider the matrix

$$A = \begin{pmatrix} 2 & 6 & -6 \\ 0 & 5 & -2 \\ 0 & 0 & 4 \end{pmatrix}.$$

a) Find its eigenvalues. Is it possible to find the eigenvalues without computing?
b) Is this matrix diagonalizable? Find out without computing anything.
c) If the matrix is diagonalizable, diagonalize it.

2.7. Diagonalize the matrix

$$\begin{pmatrix} 2 & 0 & 6 \\ 0 & 2 & 4 \\ 0 & 0 & 4 \end{pmatrix}.$$

2.8. Find all square roots of the matrix

$$A = \begin{pmatrix} 5 & 2 \\ -3 & 0 \end{pmatrix}$$

i.e. find all matrices B such that $B^2 = A$. **Hint:** Finding a square root of a diagonal matrix is easy. You can leave your answer as a product.

2.9. Let us recall that the famous Fibonacci sequence:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, \ldots$$

is defined as follows: we put $\varphi_0 = 0$, $\varphi_1 = 1$ and define

$$\varphi_{n+2} = \varphi_{n+1} + \varphi_n.$$

We want to find a formula for φ_n. To do this

a) Find a 2×2 matrix A such that

$$\begin{pmatrix} \varphi_{n+2} \\ \varphi_{n+1} \end{pmatrix} = A \begin{pmatrix} \varphi_{n+1} \\ \varphi_n \end{pmatrix}$$

Hint: Combine the trivial equation $\varphi_{n+1} = \varphi_{n+1}$ with the Fibonacci relation $\varphi_{n+2} = \varphi_{n+1} + \varphi_n$.

b) Diagonalize A and find a formula for A^n.

c) Noticing that

$$\begin{pmatrix} \varphi_{n+1} \\ \varphi_n \end{pmatrix} = A^n \begin{pmatrix} \varphi_1 \\ \varphi_0 \end{pmatrix} = A^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

find a formula for φ_n. (You will need to compute an inverse and perform multiplication here).

d) Show that the vector $(\varphi_{n+1}/\varphi_n, 1)^T$ converges to an eigenvector of A. What do you think, is it a coincidence?

2.10. Let A be a 5×5 matrix with 3 eigenvalues (not counting multiplicities). Suppose we know that one eigenspace is three-dimensional. Can you say if A is diagonalizable?

2.11. Give an example of a 3×3 matrix which cannot be diagonalized. After you constructed the matrix, can you make it "generic", so no special structure of the matrix could be seen?

2.12. Let a non-zero matrix A satisfy $A^5 = \mathbf{0}$. Prove that A cannot be diagonalized. More generally, any non-zero nilpotent matrix, i.e. a non-zero matrix satisfying $A^N = \mathbf{0}$ for some N cannot be diagonalized.

2.13. Eigenvalues of a transposition:

a) Consider the transformation T in the space $M_{2\times 2}$ of 2×2 matrices, $T(A) = A^T$. Find all its eigenvalues and eigenvectors. Is it possible to diagonalize this transformation? **Hint:** While it is possible to write a matrix of this linear transformation in some basis, compute characteristic polynomial, and so on, it is easier to find eigenvalues and eigenvectors directly from the definition.

b) Can you do the same problem but in the space of $n \times n$ matrices?

2.14. Prove that two subspaces V_1 and V_2 are linearly independent if and only if $V_1 \cap V_2 = \{\mathbf{0}\}$.

Supplementary problems for HW5

I. a) Give an example of a pair of matrices in $M_{2\times 2}^{\mathbb{Z}/5\mathbb{Z}}$ that have the same determinant but are not similar.

b) Give two examples of matrices in $M_{2\times 2}^{\mathbb{Z}/5\mathbb{Z}}$ that are not diagonalizable; or explain why all of these matrices are diagonalizable.

II. Of the 5^4 matrices in $M_{2\times 2}^{\mathbb{Z}/5\mathbb{Z}}$, how many are diagonalizable, and how many aren't?

III. We saw in class that the limiting ratio of consecutive terms of the Fibonacci sequence (with the recursive rule $F_n = F_{n-1} + F_{n-2}$) is $\frac{1+\sqrt{5}}{2}$. Come up with a different (linear) recursive rule, find its corresponding linear transformation, and calculate limiting ratio of consecutive terms.

IV. Let A and B be $n \times n$ complex matrices and assume that they commute: $AB = BA$. Prove that the matrices A and B share at least one eigenvector in common (but not necessarily with the same eigenvalue for the eigenvector).

V. As on HW4, consider again the set $M_{0,1}(n)$ of $n \times n$ matrices that have 0 or 1 for each of their entries. The set $M_{0,1}(n)$ contains exactly $2^{(n^2)}$ matrices. You are welcome to try computer experiments as you explore the following problems.

i) What is the biggest eigenvalue that a matrix in $M_{0,1}(2)$ can have? How many matrices are there in $M_{0,1}(2)$ that have this value as an eigenvalue? When this eigenvalue occurs, how large can its algebraic multiplicity be?

ii) Same questions, but for $M_{0,1}(3)$?

iii) What can you say about the biggest possible eigenvalue for matrices in $M_{0,1}(n)$ for arbitrary n, as well as the number of matrices in $M_{0,1}(n)$ that have this as an eigenvalue?